

Shengwei Liu

Shenzhen, China | +86 177 7079 2068 | willsl050127@gmail.com
https://sl3597.github.io/

EDUCATION

Cornell Tech, Cornell University	New York, USA
MEng in Electrical and Computer Engineering , GPA: 4.038/4.0 (Top 1%), Merit Scholarship	May 2026
University College London (UCL)	London, UK
BEng in Electronic and Electrical Engineering , First-Class Honors (UK Highest Classification)	June 2025

TECHNICAL SKILLS

Programming: Python, C/C++, CUDA, Triton, Embedded C, SystemVerilog, MATLAB

Inference & Deployment: ONNX Runtime, TensorFlow Lite, TensorRT-LLM, vLLM, Triton Inference Server

AI Frameworks: PyTorch, TensorFlow, Hugging Face, scikit-learn, NumPy/Pandas

Cloud & Infrastructure: Docker, Kubernetes, Linux

EDA Tools: Synopsys (Design Compiler, PrimeTime), Cadence (Virtuoso, Innovus), Siemens EDA (QuestaSim, Calibre)

EXPERIENCE

Zhuoyu Technology (DJI Automotive), Shenzhen, China Jul. 2026 – Present
High-Performance Computing Engineer

- Built and maintained Dockerized model compilation and deployment environments, managing images and dependency versions for reproducible builds.
- Delivered the end-to-end pipeline converting ONNX models into deployable .so artifacts for in-vehicle heterogeneous compute platforms.
- Profiled deployed artifacts to collect operator-level latency data; established an output-consistency workflow between original ONNX models and deployed artifacts to meet mass-production safety requirements.

Cornell University, Computer System Lab, NY, USA Dec. 2025 – May 2026
Research Assistant Advisor: Prof. Mohamed Abdelfattah

- Reproduced KV-cache quantization framework, validating its memory–accuracy trade-offs and integrating it with LLaMa-3.1 models for efficient long-context inference.
- Built a precision benchmark framework comparing FP16, FP8, INT8, FP4, and INT4 quantization across LLM (Qwen2.5-1.5B) and VLM (Qwen2-VL-2B) models using custom nn.Linear replacement modules.
- Investigated low-bit quantization (W2A4/W2A2) for VLMs, analyzing calibration sensitivity, modality-specific quantization gaps, and mixed-precision strategies toward aggressive compression for edge deployment.
- Extended the quantization analysis to diffusion models, comparing memory footprint, latency, and FID across precisions.

Tsinghua University, HAS Lab, Beijing, China May 2025 – Sept. 2025
Research Assistant Advisor: Prof. Shuwen Deng

- Implemented a YCSB-based benchmarking framework to systematically compare CXL-enabled memory systems against Redis cluster deployments, focusing on latency, throughput, and scalability under high-concurrency workloads.
- Analyzed performance implications of memory disaggregation and interconnect topologies in distributed KV stores, focusing on architectural trade-offs under varying concurrency levels.
- Evaluated Paxos/Raft variants optimized for CXL shared memory and RDMA, reducing replication latency by bypassing kernel networking stacks.

PROJECTS

GPU LLM Inference Optimization (Python, PyTorch, Triton) Feb. 2026 – Mar. 2026

- Optimized a custom sparse-attention transformer inference pipeline on GPU, achieving 2.5× decode throughput and 4.4× prefill speedup over baseline across four evaluation regimes.
- Implemented custom Triton kernels for block-sparse attention with dynamic block skipping and fused RMSNorm, eliminating redundant global memory loads and reducing kernel launch overhead.
- Removed GQA key/value expansion in the decode path by handling multi-query grouping directly inside the Triton attention kernel, cutting memory bandwidth usage for decode by 7×.

Embedded Keyword Spotting & Model Compression (Python, Embedded C) Feb. 2026 – Mar. 2026

- Trained a TinyConv model on Google Speech Commands, reaching 93.6% test accuracy.
- Deployed via TFLite to an Arduino Nano 33 BLE, achieving ~110 ms real-time inference latency.
- Applied PTQ and QAT, retaining near-lossless accuracy under INT8 (~93.2%).
- Applied structured pruning, reducing FLOPs by up to 8× and on-device latency by ~28%.

Hand Motion Detection using IMU and PPG Sensors (Python, Embedded C) Oct. 2024 – Apr. 2025

- Developed a real-time hand motion detection system integrating 6-axis Inertial Measurement Unit and 4-channel Photoplethysmography sensors, enabling Bluetooth-based multi-modal data streaming and edge-side preprocessing.
- Collected 1,800 samples across 6 hand motions, each with 15-time steps and 10 fused features, and utilized SciPy, Keras, and Scikit-learn for preprocessing and feature extraction, including Min-Max normalization, signal filtering, and stratified splitting.
- Trained a stacked LSTM network (25-15-10 units) with dropout and batch normalization via TensorFlow, achieving 94.05% test accuracy and 0.94 macro F1-score, outperforming SVM, KNN, and Random Forest baselines.
- Deployed LSTM models on an Arduino Nano 33 IoT using the LiteRT framework and implemented a Python-based BLE inference pipeline with the Bleak package for real-time motion feedback to a PC without data loss.

PUBLICATION

Shengwei Liu, et al. “SHMemora: Protective Key–Value Store on Distributed Shared Memory”. In: *Proceedings of the 42nd IEEE International Conference on Data Engineering (ICDE’26)*, May 2026